

Design variables

$$\min_x f(x)$$

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$x = \{0, 1, 2, \dots\}$$

- Design variables can be continuous or discrete, or a mixture.

$$R = \{10, 27, 47, 100, \dots\}$$

- Design variables should be independent, otherwise there may be an infinite number of equivalent solutions.

$$\min_x (x_1 + x_2)^2$$

Solution $x_1 + x_2 = 0$

- Don't forget to add physically meaningful bounds.

$$\underline{x_i} \leq x_i \leq \overline{x_i}$$

lower bound upper bound

- Tip: when starting to solve a problem, try to use as few design variables as possible.

Objective function (cost function)

- This is the objective by which we measure the quality of a solution.
- In this course, we will usually have a closed-form function $f(x)$ which serves as the objective to be minimised or maximised.
e.g. $f(x) = x_1^2 + 3x_1x_2 + x_2^2$
- However, in many applications, the objective function could be the output of a simulation or physical system/experiment.

$\min_x f(x)$
↑
obj.

function

converts
vector x
 $n \times 1$
to scalar
number

Objective function

$$\max_x f(x) \iff \min_x -f(x)$$

- We will look at minimisation problems in this course. If you have a maximization problem, you can always simply reverse the sign of the objective function!
- The choice of objective function is crucial for successful design optimization. If this is done poorly, the mathematical optimum will be non-optimal from the engineering point of view.
- A bad choice for the objective function is a common mistake in design optimization.

Constraints

$$\min_x f(x)$$

$$\text{s.t. } \left. \begin{array}{l} x_1 \leq 3x_2 \\ x_2 = 0 \end{array} \right\} \text{constraints}$$

- Many optimisation problems involve constraints:
 - E.g. when designing a car you are limited by the available parts, by road rules governing size, by safety requirements etc.
 - When grocery shopping you are limited by stock availability, budget limits, item limits, etc.

$$g(x) = x_1 - 3x_2 \leq 0$$

- There are two types of constraints:

- Inequality constraints

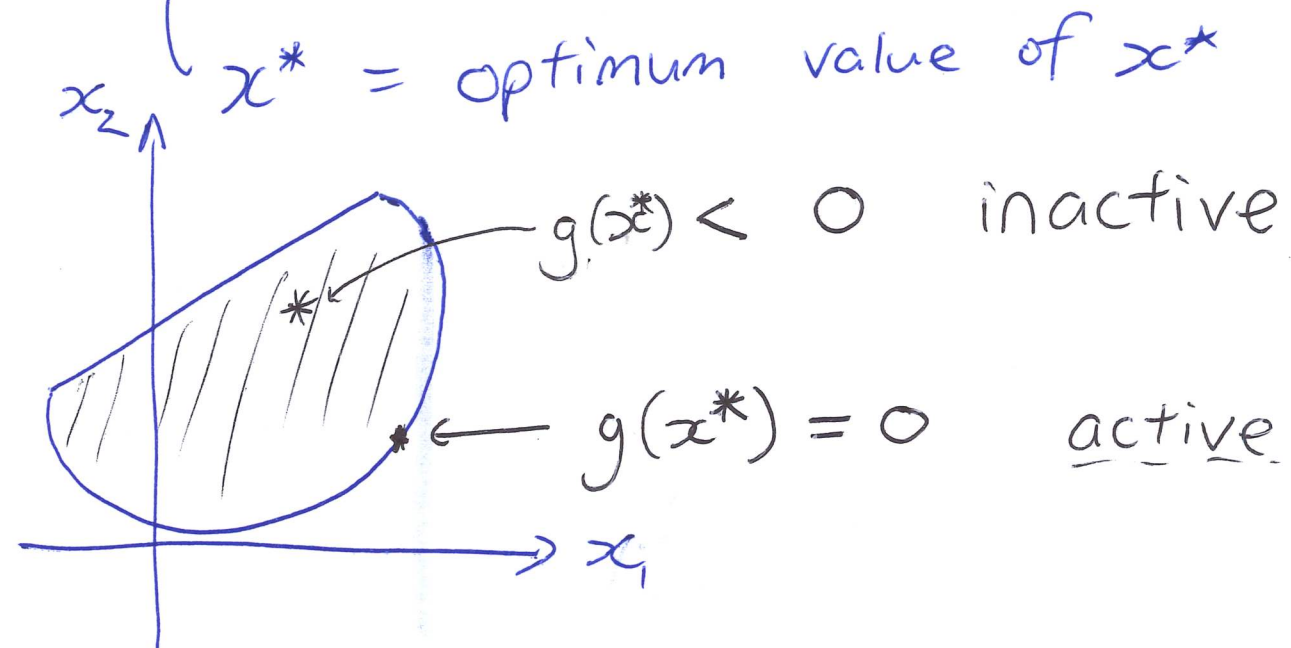
$$g(x) \leq 0$$

- Equality constraints (equivalent to two inequality constraints)

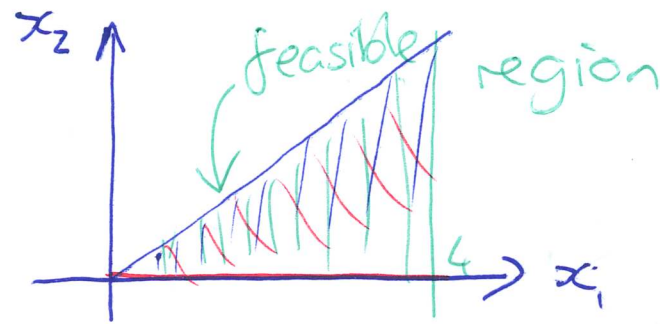
$$h(x) = x_2 = 0 \Leftrightarrow \begin{cases} x_2 \geq 0 \\ x_2 \leq 0 \end{cases}$$

Constraints

- Inequality constraints can be active or inactive at the optimum point. An active inequality constraint means that $g(x^*) = 0$, whereas for an inactive one, $g(x^*) < 0$.



Constraints



$$x_1 < x_2$$

$$x_2 \geq 0$$

$$x_1 \leq 4$$

- The region in which all constraints are satisfied is called the feasible region.
- If there are too many constraints, there may be no feasible region and this makes the problem also *infeasible*.
- Example: if there are more independent equality constraints than design variables, the problem is infeasible.

Putting it all together (+notation)

$$\begin{array}{ll} \text{design} & \text{objective} \\ \text{variable} & \downarrow \\ \min & f(\mathbf{x}) \\ \rightarrow \mathbf{x} & \swarrow \\ \text{subject to } \underline{x}_i \leq x_i \leq \bar{x}_i, i = 1, 2, \dots, n & \text{number of design variable bounds} \\ & \downarrow \\ g_j(\mathbf{x}) \leq 0, j = 1, 2, \dots, n_g & \text{number of inequality constraint} \\ h_l(\mathbf{x}) = 0, l = 1, 2, \dots, n_h. & \text{equality} \end{array}$$

Notation

- Vectors

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

x_0 - starting point
 $n \times 1$

- Matrices and elements of matrices

$$\begin{bmatrix} A_{11} & A_{1n} \\ A_{n1} & A_{nn} \end{bmatrix}$$

- Optimum point x^*

Vector and matrix norms

$$\|x\| \geq 0$$

norm

$$\|x\| = 0 \quad \text{iff} \quad x = 0$$

$$\|x\|_2 = \left(\sum_{i=1}^n x_i^2 \right)^{1/2} = (x_1^2 + x_2^2 + \dots + x_n^2)^{1/2}$$

$$\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p} = (|x_1|^p + |x_2|^p + \dots + |x_n|^p)^{1/p}$$

$p=1$: absolute value sum

$p=\infty$: maximum absolute value in vector $x = \begin{bmatrix} 0 \\ 10 \\ -20 \end{bmatrix}$

$$\|x\|_2 = \sqrt{500}$$

$$\|x\|_1 = 30$$

$$\|x\|_\infty = 20$$

Matrices

diagonal

$$\begin{bmatrix} * & & & \\ & * & & \\ & & * & \\ 0 & & & \ddots \\ & & & & * \end{bmatrix}$$

$$A_{ij} = 0 \text{ if } i \neq j$$

identity

$$I_n = \begin{bmatrix} 1 & & \\ & 1 & \\ & & \ddots \\ 0 & & & 1 \end{bmatrix}$$

norm

$$\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^m |A_{ij}|$$

max absolute
column sum

$$\|A\|_2 = (\lambda_{\max}(A^T A))^{1/2}$$

$$\|A\|_{\infty} = \max_{1 \leq i \leq m} \sum_{j=1}^n |A_{ij}|$$

row

$$\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n A_{ij}^2}$$

Taylor Series

$$f(x) \approx f(x_0)$$

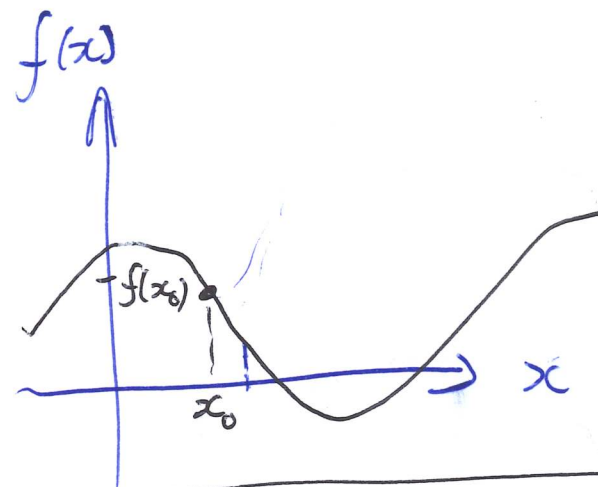
$$+ \underbrace{f'(x_0)}_{\text{slope}} (x - x_0)$$

$$+ \frac{1}{2} f''(x_0) (x - x_0)^2$$

$$+ O(x - x_0)^3$$

$$J(x) = \frac{\partial}{\partial x} f(x) =$$

$$\begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$$



Hessian

$$H(x) = \frac{\partial}{\partial x_i} \frac{\partial}{\partial x_j} f(x)$$

$$\begin{bmatrix} \frac{\partial^2 f(x)}{\partial x_1^2} & \frac{\partial^2 f(x)}{\partial x_1 \partial x_2} & \dots \\ \frac{\partial^2 f(x)}{\partial x_2 \partial x_1} & \ddots & \\ \vdots & \ddots & \frac{\partial^2 f(x)}{\partial x_n^2} \end{bmatrix}$$